

BrandGuard AI Capstone Project Report

AI-Powered Brand-Safe Marketing Content Generation System

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Name of the project- Brand Safe Content and campaign studio.

Abstract

BrandGuard AI is a workflow automation solution built with n8n, OpenAI and Google Sheets. It automates marketing content generation using approved product information and brand guidelines while validating outputs and supporting human review.

Introduction

This project demonstrates the integration of Generative AI with workflow automation to create brand-compliant marketing content efficiently.

Problem Statement

Organizations invest considerable time and effort in producing marketing content that complies with internal brand policies. Manual review processes are slow, repetitive, and susceptible to human error. There is a need for an intelligent system that generates high-quality marketing content while ensuring compliance with approved product information and organizational branding standards before publication.

Objectives

- * Develop an AI-powered content generation workflow.
- * Read approved product information from Google Sheets.
- * Read company brand guidelines.
- * Generate professional marketing content using OpenAI.

- * Validate that the generated content is not empty.
- * Store generated content automatically.
- * Support human review before publishing.
- * Demonstrate workflow automation using n8n.

Technology Stack

n8n Cloud

OpenAI API

Google Sheets

AI Agent

IF Node

System Architecture

User Starts Workflow

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Manual Trigger

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Google Sheets

(Product Database)

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Google Sheets

(Brand Guidelines)

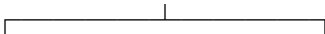


AI Agent

(OpenAI)



Validation (IF Node)



True False



Append Row Stop Workflow



Generated_Content Sheet



Pending Human Approval

Implementation

Describe each node:

1. Manual Trigger
2. Google Sheets (Products)
3. Google Sheets (Brand Guidelines)
4. AI Agent
5. OpenAI Chat Model
6. IF Validation
7. Google Sheets Append Row

[Insert workflow screenshots after each subsection.]

Testing

Test Case 1

Input:

SmartWatch X

Result:

Instagram marketing content successfully generated.

Validation:

Passed

Status:

Pending

Results

The workflow successfully generated marketing content, validated responses and stored outputs with Pending approval.

Future Scope

Email approvals, Slack integration, CRM integration, multilingual support and analytics dashboard.

Conclusion

The project demonstrates responsible AI through Human-in-the-Loop approval while reducing manual effort and improving consistency.

Screenshot Placeholders

Figure 1: Complete n8n Workflow

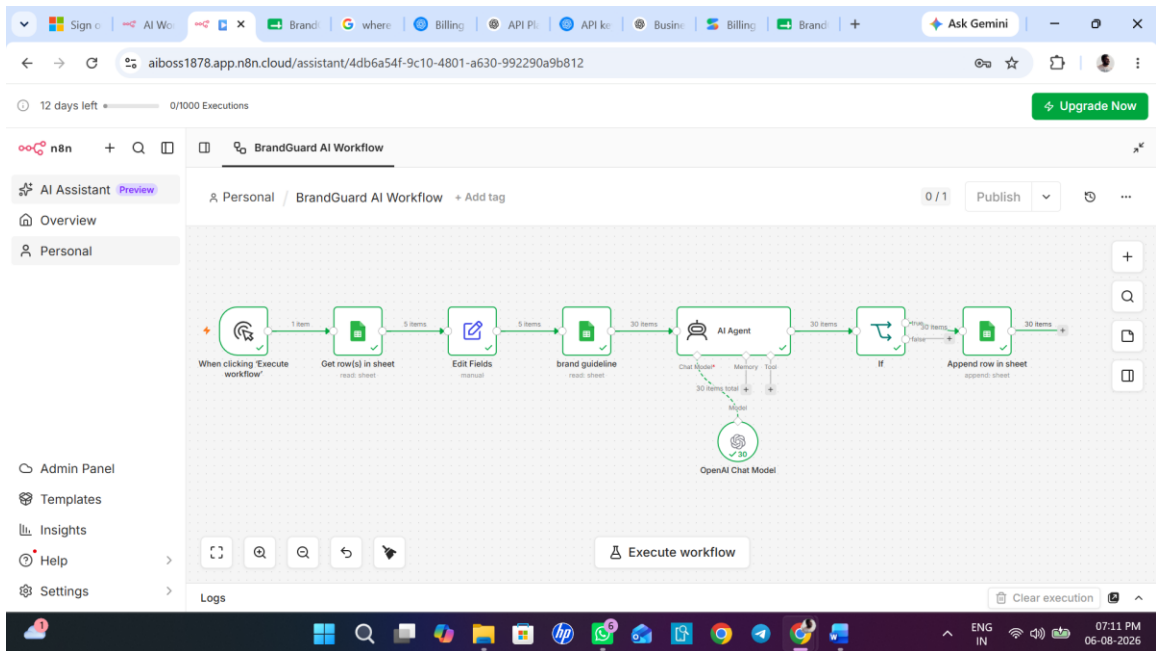


Figure 2: AI Agent Configuration

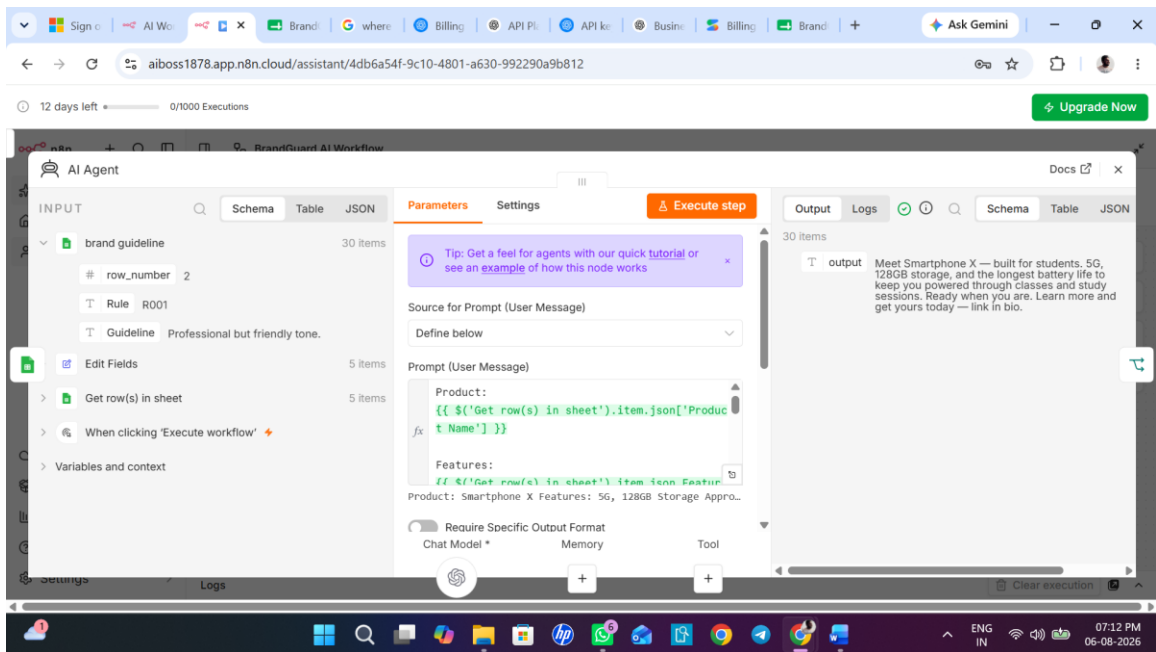


Figure 3: IF Node

Access link to google sheets -

<https://docs.google.com/spreadsheets/d/1XmR4mi8CoNGct55Q3YCtpvdPu-wA4azPBeYOQC0S1FQ/edit?usp=sharing>